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BCACAEN 601

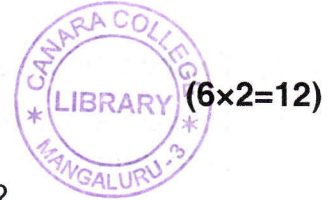
**Sixth Semester B.C.A. Degree Examination, June/July 2024
(NEP 2020) (2023 – 24 Batch Onwards)
FUNDAMENTALS OF DATA SCIENCE**

Time : 2 Hours

Max. Marks : 60

Note : Answer **any six** questions from Part – A and **one full** question from **each** Unit in Part – B.

PART – A



1. a) What is Data Mining ?
b) What is supervised learning and unsupervised learning ?
c) Define concept hierarchy. Give an example.
d) What is Data Mart ? List its types.
e) What are frequent patterns ? What is the use of frequent pattern mining ?
f) What are categorical attributes and quantitative attributes ?
g) What is Decision Tree ?
h) Expand CLARA and ROCK.

PART – B

Unit – I

2. a) Define KDD process. Explain the different stages of KDD.
b) Explain various application areas of Data Mining. (6+6)
3. a) Explain the following Data Mining techniques :
i) Verification model
ii) Discovery model.
b) List and explain the issues and challenges in Data Mining. (6+6)

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Unit – II

4. a) Define and explain data warehouse.
b) Explain any four OLAP operations.
c) Explain any two numerosity reduction techniques. (4+4+4)
5. a) Define Measure. Explain different categories of measures.
b) Explain any four ways of handling missing values.
c) Explain the different steps involved in data transformation. (4+4+4)



Unit – III

6. a) Explain the classification of frequent pattern mining.
b) Explain any three pruning strategies of mining closed frequent item sets. (6+6)
7. a) Explain Apriori algorithm.
b) What is constrain based mining ? Explain the constraints included in constrain based mining. (6+6)

Unit – IV

8. a) Explain any four criteria which are used for comparing classification and prediction methods.
b) Explain IF-THEN rules for classification.
c) Explain DBSCAN and OPTICS. (4+4+4)
9. a) Explain any four requirements of clustering in data mining.
b) Explain K-means Partition Algorithm.
c) Explain four cases of cost function for K-mediod clustering. (4+4+4)
